EE1080 Probability, EndTerm Exam

30 April, 2026

Max. Marks: 50 **Time:** 3 hours.

Instructions

- You are welcome to answer all questions. The maximum marks will be capped at 50. That is, if your answers are evaluated to x marks then your score will be $\min\{50, x\}$.
- Please **write your roll number**, and **name** prominently in the first page of the answer sheet.
- No laptops, mobile devices, cheat sheets allowed. You may use calculators
- Please write supporting arguments for any of the statements you make. any result proved in the lectures can be used by stating it clearly, no need to prove them in your answer.
- Highlight your final answer clearly for each part by drawing a box around it.
- All the best!

$\mathbb{R}$ is the set of real numbers
$\mathbb{N}$ is the set of natural numbers

1. *(8 marks)* Let X and Y be independent exponentially distributed random variables with the same parameter λ i.e.,

$$f_X(x) = f_Y(x) = \lambda e^{-\lambda x}, \ x \geq 0.$$

Given the following four functions of X and Y :

$$U = \max(X, Y), \ V = \min(X, Y), \text{ and } W = U - V, Z = X + Y.$$

(a) Find the joint pdf of U and V. Are they independent?

(b) Find the joint pdf of V and W. Are they independent?

(c) Find the joint pdf of Z and W. Are they independent?

(d) Find the pdf of Z.

2. *(11 marks) Moment Generating Functions (MGF)*

(a) (2) Let $X \sim N(\mu, \sigma^2)$. Find $M_X(s)$. Is it well defined for every $s \in \mathbb{R}$?

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

(b) (2) Suppose $\mu = 0, \sigma = 1$, find the k-th moment of X, $E[X^k]$ for $k \in \mathbb{N}$. (*Hint: Use Taylor series expansion of the $M_X(s)$.*)

(c) (2) In continuation to earlier problem, suppose $Y = a + bX + cX^2$. Find correlation between X, Y i.e., find $\rho(X, Y)$.

(d) (3) Find MGF of $Y = X_1^2 + X_2^2 + \cdots + X_n^2$ where X_i are i.i.d standard normal random variables. Is it well defined for every $s \in \mathbb{R}$.

(e) (2) Let $\Psi(s) = log(M_X(s))$. Prove/Disprove the following statement:

$$Var(X) = \Psi''(t)|_{t=0}.$$

3. *(5 marks)* Suppose you are given a sample of uniform random variable $U \sim$ Uniform([0, 1]). How would you

(a) (1.5) generate sample of a Geometric(0.5) random variable.

(b) (1.5) generate sample of an Exponential(λ) random variable?

(c) (2) Suppose you have two i.i.d uniform random variable samples U_1, U_2. How do you generate a sample of a random variable that has p.d.f $f_X(x) = \lambda^2 x e^{-\lambda x}$ for $x > 0$. (*Hint: identify how distribution of X is connected to exponential random variables.*)

4. *(14 marks)* Let X be a random variable representing the precision (inverse variance) of a sensor. Suppose our prior knowledge about X follows the distribution shown below:

$$f_X(x) = \lambda^2 x e^{-\lambda x} \text{ for } x \geq 0, \lambda = 10.$$

Given a specific precision $X = x$, we take an measurement, Y where $Y \sim N(0, 1/x)$.

(a) (1) Describe $f_{Y|X}(y|x)$ as a function of x, y.

(b) (3) Find $f_{X|Y}(x|y)$. Does it follow Gamma distribution ? Describe the parameters (α, β) of the Gamma distribution if true. *Hints: You don't have to find the exact $f(y)$ to find the conditional density assume the conditional pdf as a function of x then $f(y)$ is simply a constant. Description of the Gamma distribution and its parameters are provided below :*

i. *If $Z \sim$* Gamma(α, β) *then:*

$$f_Z(z) = \frac{\beta^\alpha}{\Gamma(\alpha)} z^{\alpha-1} e^{-z\beta}, \quad z \geq 0.$$

ii. $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ *and* $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(1) = 1$.

iii. $E[Z] = \frac{\alpha}{\beta}$, $Var(Z) = \frac{\alpha}{\beta^2}$

(c) (2+2) Find the MAP and ML estimates of X from $Y = y$

(d) (2) Find the MMSE estimate

(e) (1) By staring at the MMSE estimator can you tell if the linear MMSE estimator is the same as MMSE estimator ?

(f) (2) Find the mean square error of the LMMSE estimate *Hint: Use total law of expectation*

(g) (1) Which estimate out of MAP/ML/MMSE results in smallest mean square error?

5. (*4 marks*) Let $X_1, \cdots, X_n$ be i.i.d. exponential(λ) random variable with $\lambda = 1$. Let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

(a) (2) Using Chebyshev's inequality, report the smallest value of n such that

$$P(0.9 \leq S_n \leq 1.1) \geq 0.99.$$

(b) (2) Use central limit theorem to approximate $P(0.9 \leq S_n \leq 1.1)$ in terms of $\phi(x)$ where:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} dt$$

6. (*7 marks*) *Convergence of Random Variables:* Let X_n's be independent random variables such that $X_n \sim N(0, \frac{1}{n})$.

(a) (1) Does X_n converge in distribution sense to 0 ?

(b) (2) Does X_n converge in probability sense to 0 ?

(c) (3) Does X_n converge in almost sure sense to 0 ?

(d) (1) Does X_n converge in mean square sense to 0 ?

7. (*7 marks*) *Gaussian Random Vectors:* Let $X = (X_1, X_2, X_3)^T$ be a Gaussian vector with covariance matrix: and mean vector described as:

$$K_X = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 16 \end{bmatrix}, E[X] = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$$

(a) (3) Given samples of three i.i.d standard normal random variables. $W = (W_1, W_2, W_3)^T$ describe how you would generate samples of the random vector X.

(b) (4) Suppose you restrict your attention to the random vector $\hat{X} = (X_1, X_2)$, describe and plot the points that evaluate to 90 percent of the maximum possible p.d.f evaluation of $f_{\hat{X}}(x_1, x_2)$.